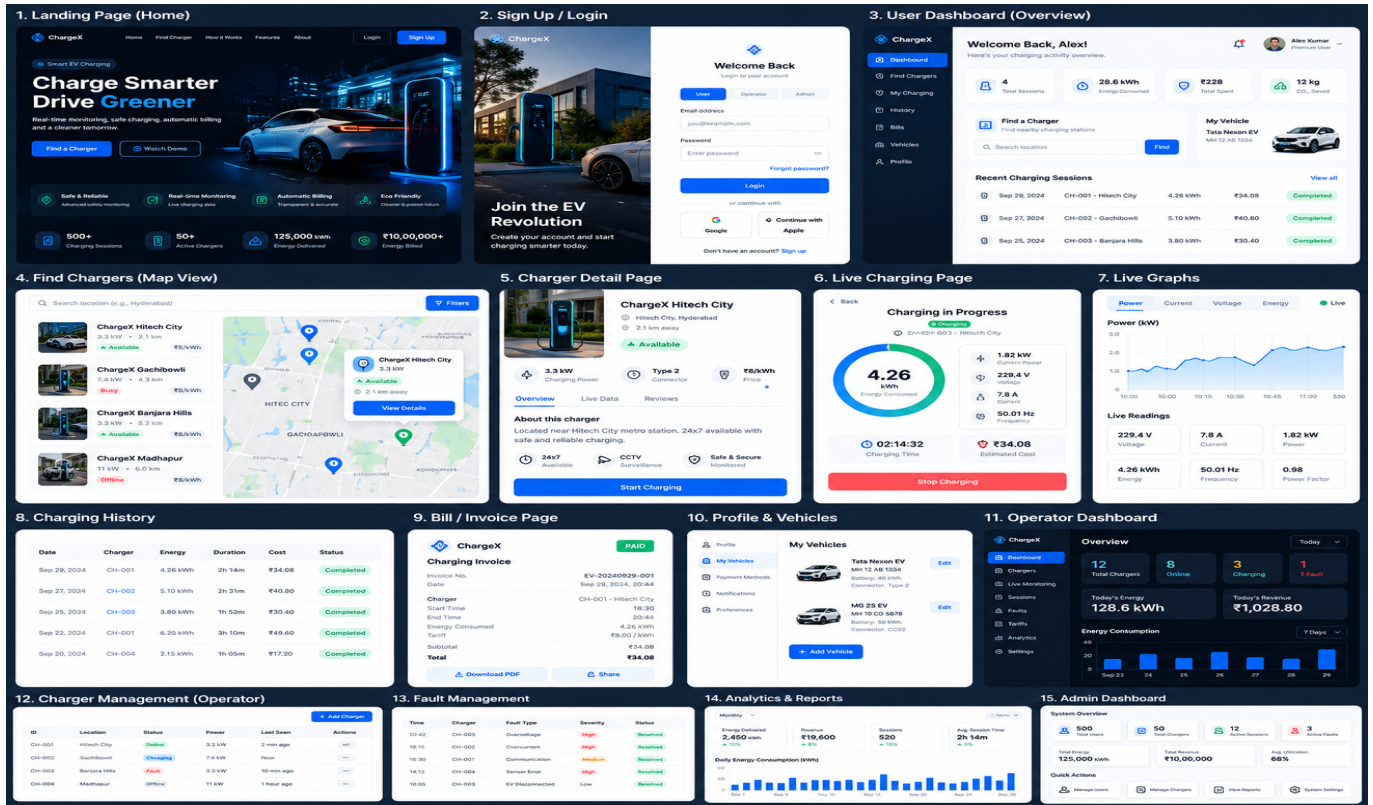


SMART EV CHARGING PLATFORM

Website • UX/UI • IoT • Payments • Backend • Database • Developer Handoff

Architecture & Product Blueprint



Prepared from the uploaded project proposal and the agreed product/engineering design extensions.

1. Executive Summary

The proposed website is the digital platform for an IoT-based smart EV charging system. The source proposal describes a single-phase 230 V, 50 Hz EV charging system with real-time monitoring, regulated charging, PFC, automatic billing, EV connection detection, local display, data storage and IoT-based remote monitoring.

- Build a complete smart-charging platform rather than only a sensor dashboard.
- Support charger discovery, charging sessions, energy/cost visibility, history, bills, wallet and RFID management.
- Support two charging authorization/payment experiences: Scan & Charge using QR/UPI and Tap & Go using RFID.
- Provide operator/admin tools for charger monitoring, faults, tariffs, payments and analytics.
- Keep safety-critical cutoff local to the charger/controller.

2. Requirements Derived from the Original Project

Requirement	Proposal basis	Website implication
Regulated charging	Stable and efficient charging	Show charger state and readiness
Real-time monitoring	Voltage, current and power	Live telemetry cards and graphs
Power quality	PFC and reduced harmonics	Power-quality screen for measured metrics
Automatic billing	Energy × tariff	Live cost and final bill
EV detection	Connection/start/end time	Charging session lifecycle
Local display	V/I/P/energy/time/bill	Mirror key operational metrics
Data storage	Users, history, energy, billing	PostgreSQL
IoT communication	Remote monitoring	MQTT + WebSocket
Fault safety	Relay cutoff and alarms	Fault center; local cutoff remains primary

Important scope clarification

The source proposal is inconsistent on electrical configuration: the introduction says single-phase 230 V, 50 Hz, while the methodology diagram shows 230 V / 415 V. The final hardware specification must explicitly resolve this before implementation.

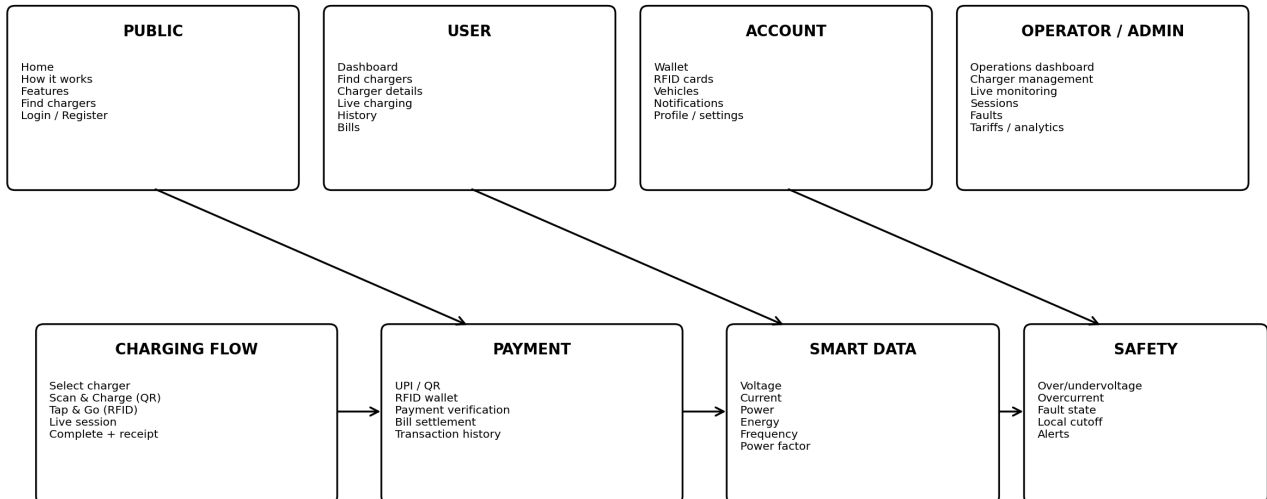
3. Product Vision

Position the website as a Smart EV Charging Management Platform. Core journey:

FIND → AUTHORIZE → CHARGE → MONITOR → CALCULATE → PAY → REMEMBER

4. Website Sitemap

WEBSITE INFORMATION ARCHITECTURE / SITEMAP



Area	Recommended pages	Purpose
Public	Home, How It Works, Features, Find Chargers, Login/Register	Public discovery and conversion
User	Dashboard, Charger Details, Live Charging, History, Bills, Wallet, RFID, Vehicles/Profile	Complete charging journey
Operator/Admin	Operations, Chargers, Live Monitoring, Sessions, Faults, Tariffs, Payments, Analytics, Users, Settings	Fleet and platform management

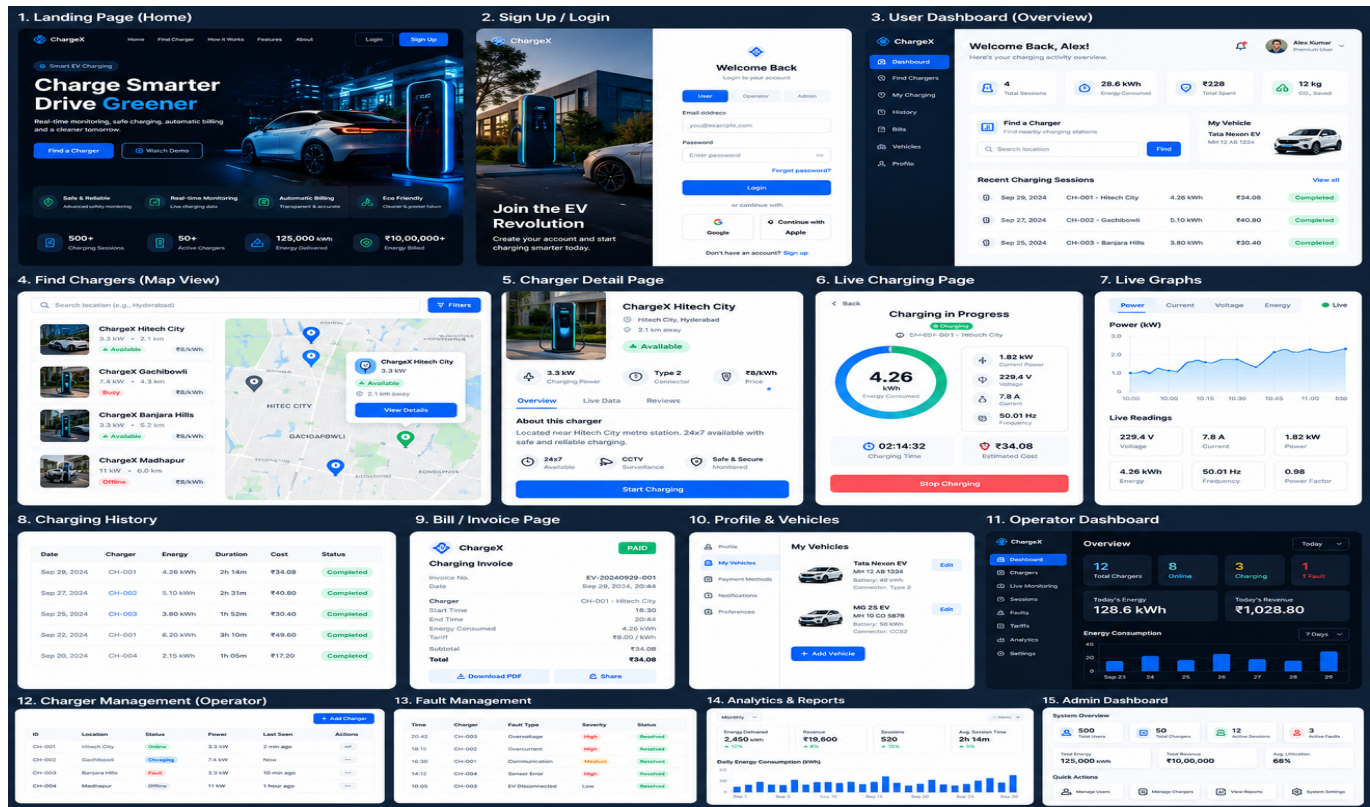
5. Screen-by-Screen Blueprint

Page	Purpose	Primary action
Home	Premium hero, benefits, how it works, safety and CTA	Find a Charger
Login/Register	Secure user access	Login/Create
Dashboard	Active session, wallet, recent sessions and quick actions	Find Charger
Find Chargers	Map/list, search, availability, distance, power, price	View Details
Charger Details	Availability, power, connector, price, safety status	Start Charging
Start Charging	Choose QR/UPI or RFID	Authorize
Live Charging	Power, voltage, current, energy, time, cost	Stop
Charging Complete	Final energy, duration, bill and payment status	View Bill
History	Past sessions and payment method	Open Session
Bill	Invoice and transaction details	Download/Share
Wallet	Balance and transactions	Add Money
RFID Cards	Active/blocked cards and management	Manage
Vehicles	Saved EV information	Edit
Notifications	Charging/payment/fault alerts	Open Event
Profile	Account and settings	Save
Operator Dashboard	Fleet KPIs, chargers, energy, revenue, faults	Open Charger
Charger Management	Manage devices and status	Manage

Page	Purpose	Primary action
Live Monitoring	Real-time telemetry and power quality	Inspect
Faults	Faults, severity, state, resolution	Investigate
Analytics	Energy, revenue, sessions, utilization, quality	Filter/Export
Tariffs	Configure energy pricing	Update
Payments	Payment and settlement records	Inspect

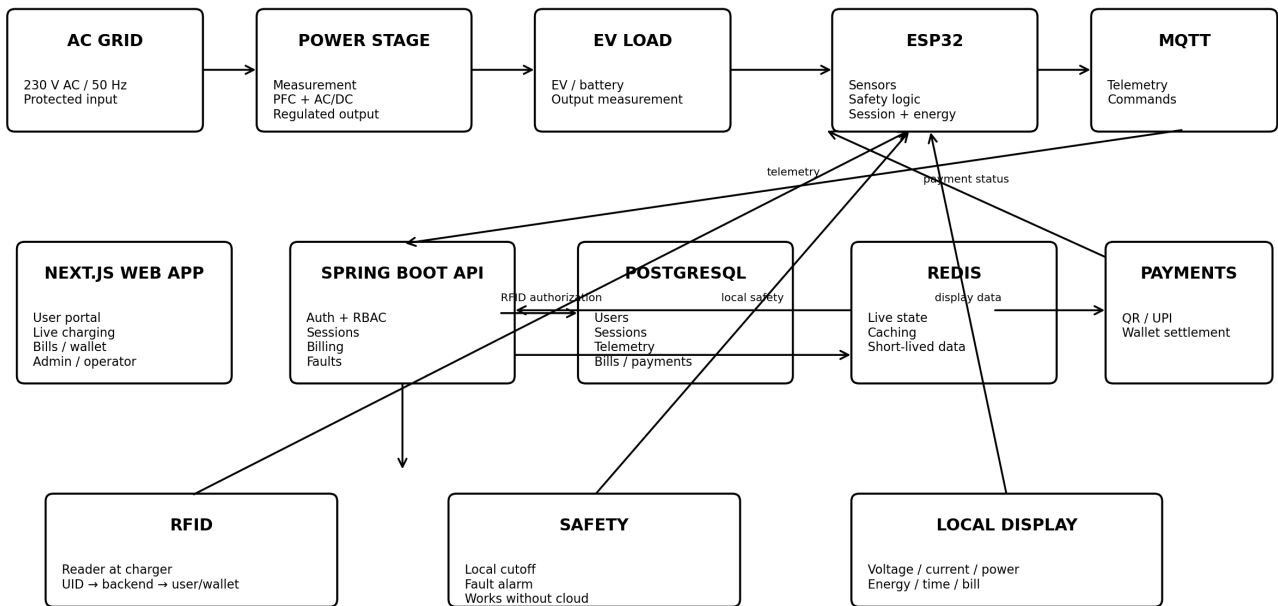
6. Modern UI/UX Direction

- Dark navy/deep blue foundation with electric-blue accents.
- Rounded cards, clear typography, restrained shadows and subtle motion.
- Strong visual hierarchy; one primary metric per screen.
- Responsive desktop/mobile layouts.
- Consistent status colors and clear offline/error/loading states.
- Use 'Scan & Charge' for QR and 'Tap & Go' for RFID.



7. Technical Architecture

SMART EV CHARGING PLATFORM — SYSTEM ARCHITECTURE



Recommended separation: Next.js for presentation, Spring Boot for business logic and authorization, PostgreSQL for durable relational data, Redis for live state/caching, MQTT for device telemetry, WebSocket for browser updates, and a payment provider for QR/UPI.

Safety principle: the charger/controller must stop charging locally when required, even if the website or internet is unavailable.

8. Technology Stack

Layer	Technology	Why
Frontend	Next.js + TypeScript	Structured routing and maintainable product UI
UI	Tailwind CSS + shadcn/ui	Consistent modern components
Charts	Recharts	Energy/power/voltage/current graphs
Backend	Spring Boot + Java	Business APIs and strong fit for the project
API	REST	Users, chargers, sessions, billing and admin
Realtime	WebSocket	Live browser updates
IoT	MQTT	Lightweight device telemetry
Database	PostgreSQL	Relational integrity
Cache	Redis	Live state and caching
Hardware	ESP32	Wi-Fi and sensor/control integration
Auth	JWT + refresh + RBAC	Secure role-based access
Deployment	Docker + GitHub Actions	Repeatable builds and CI/CD

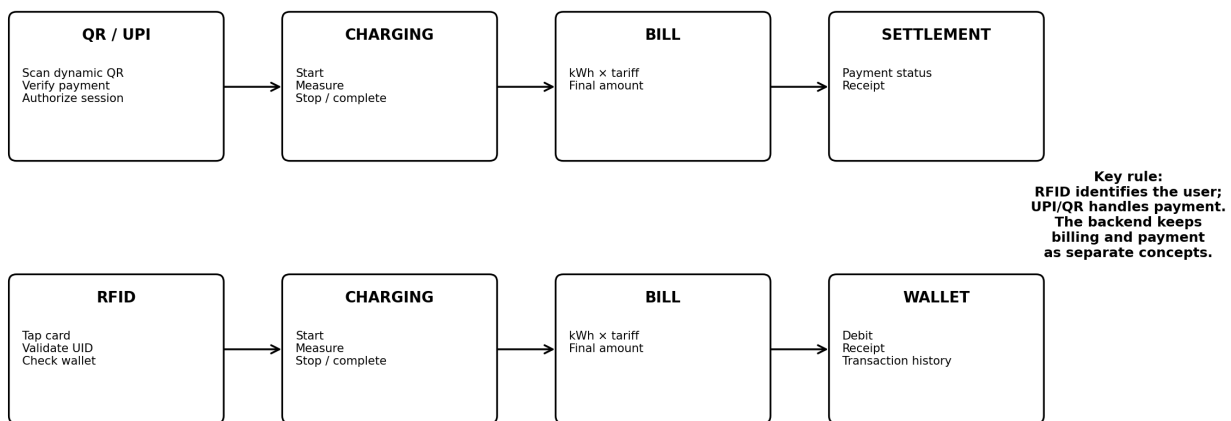
9. Database Blueprint

Entity	Responsibility
users	Identity, role, status
vehicles	Vehicle information
chargers	Device identity, location, power, status
charging_sessions	Session lifecycle and energy

Entity	Responsibility
measurements	Time-series electrical values
tariffs	Energy price configuration
bills	Consumption-based charge
payments	Payment method, provider reference, status
wallets	RFID-linked balance
wallet_transactions	Credits/debits/refunds
rfid_cards	Card UID, user association, status
faults	Safety/communication fault events
notifications	User alerts

10. QR/UPI and RFID

PAYMENT & CHARGING FLOWS



QR/UPI — Scan & Charge

• Select charger → choose Scan & Charge → generate/launch payment flow → verify payment server-side → authorize session → charge → final bill/settlement → receipt.

RFID — Tap & Go

• Tap registered RFID → validate UID/card status → check account/wallet eligibility → authorize session → charge → calculate bill → settle configured wallet/account flow → receipt.

RFID should identify the user rather than store the wallet balance on the card. A lost card can then be blocked without destroying the user's account balance.

11. Real-Time IoT Flow

Sensors → ESP32 → MQTT → Spring Boot → Redis/PostgreSQL → WebSocket → Next.js

Only expose telemetry that the hardware actually measures. Demo/mock values must be clearly separated from live production data.

12. Safety & Fault Architecture

- Local cutoff and protection are primary.
- Use explicit charger states such as AVAILABLE, CONNECTED, CHARGING, COMPLETED, FAULT and OFFLINE.
- Monitor the final set of conditions approved by the electrical design.

- On confirmed fault: local cutoff/alarm, log event, then report to backend when communication is available.

13. Suggested API Surface

Area	Example endpoints
Auth	POST /api/auth/login • /refresh • /register
Chargers	GET /api/chargers • GET /api/chargers/{id} • /status
Sessions	POST /api/charging-sessions • GET /{id} • POST /{id}/stop
Telemetry	POST /api/measurements • GET /charging-sessions/{id}/measurements
Billing	GET /api/bills • GET /api/bills/{sessionId}
Payments	POST /api/payments/create • POST /api/payments/webhook
Wallet	GET /api/wallet • POST /api/wallet/top-up
RFID	GET/POST /api/rfid-cards • PATCH /{id}/status
Faults	GET /api/faults • GET /api/chargers/{id}/faults

14. UX Rules

- Primary action must be obvious.
- Live charging prioritizes power, energy, time and cost.
- Show safety faults clearly.
- Show last-seen time for offline chargers.
- Do not invent unsupported telemetry.
- Use short payment flows and explicit confirmation for irreversible actions.
- Support desktop and mobile.

15. Roadmap

Phase	Deliverable
1	UX prototype with mock telemetry
2	Spring Boot backend and authentication
3	PostgreSQL schema and reporting
4	MQTT + Redis + WebSocket
5	ESP32 hardware integration
6	QR/UPI payment + RFID
7	End-to-end integration and testing
8	Security, observability, backups and production hardening

16. Decisions Required Before Hardware Integration

- 230 V single-phase vs 415 V three-phase.
- Maximum charging power and current.
- EV/battery voltage range.
- Charging interface/standard.
- Exact PFC approach.
- Metering accuracy.
- Payment provider.

- RFID reader/protocol.
- Tariff model.
- Cloud/deployment target.

17. Developer Acceptance Criteria

- Role-based login and access works.
- Users can find chargers and view availability.
- QR and RFID authorization paths work.
- QR payment is verified server-side.
- RFID can authorize without the website being open.
- Sessions record start/end, energy and status.
- Live telemetry updates without refresh.
- Bills use measured energy and configured tariff.
- Payment status is separate from bill status.
- RFID cards can be blocked.
- Faults are logged and visible to operators.
- Local safety remains functional without internet.
- Responsive, loading, empty, offline and error states are implemented.

18. Engineering Notes

The source proposal is concept-level and leaves several hardware and payment decisions open. Product extensions such as maps, wallets, RFID management, notifications, operator dashboards and analytics are recommendations from this blueprint, not claims that they were already present in the original proposal.

For real mains/EV power electronics, use appropriately rated protection, isolation, switching, sensing, enclosures and qualified electrical engineering/testing practices. The website must never be treated as a safety controller.

19. Source Reference

Primary source: uploaded 9-page presentation "IOT-Based smart EV charging unit." Relevant source sections: Introduction (p.2), Literature Review (pp.3–4), Project Objectives (p.5), Methodology Block Diagram (p.6), Methodology Overview (p.7), Proposed Outcomes (p.8), References (p.9).